

AI-POWERED HEALTHCARE PLATFORM

ArogyaAI

From patient symptoms to doctor-ready clinical briefs —
in any language, before the consultation begins.

Team Name • Hackathon 2026

THE PROBLEM

Meet Ramesh

"Ramesh is 58. He carried a folder of lab reports to a government hospital.

He waited

3 hours.** His consultation lasted **90 seconds.

The doctor never opened his folder. Ramesh has undiagnosed diabetes."

This isn't rare. This is the norm.



Under 2 minutes — average consultation time in India [Irving et al., BMJ Open 2017]



Paper reports ignored — doctors have no time to read unstructured documents



Language barrier — most Indians don't speak English, yet health AI is English-only

1:1,457

Doctor-patient ratio
WHO recommends 1:1,000

~63%

Out-of-pocket
health spending

22

Official languages
Health AI speaks 1

ArogyaAI — Making Every 2 Minutes Count

An end-to-end AI platform that speaks the patient's language, understands their documents, and prepares the doctor — before the consultation begins.



Speak Your Language

Voice input in Hindi, Marathi, English — even mixed Hinglish



AI Understands Reports

Lab values extracted with AI reliability indicators you can verify

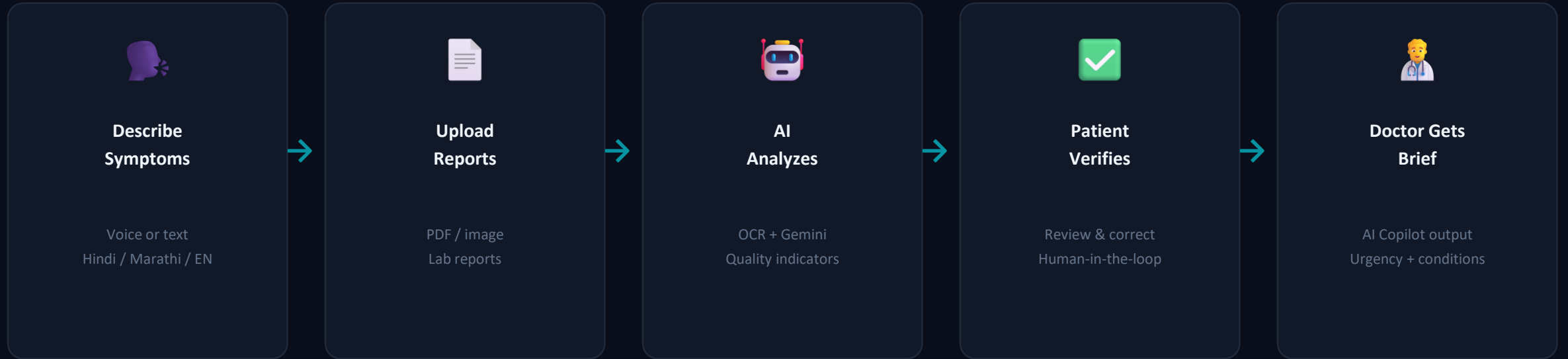


Doctor Gets a Brief

Structured clinical copilot with urgency, conditions, and focus areas

💡 Key: These aren't separate apps. It's one integrated pipeline — symptoms → documents → booking → AI copilot brief → doctor.

The Patient Journey — End to End



 Screenshot: Patient Dashboard
Capture from /dashboard

 Screenshot: Doctor Dashboard
Capture from /doctor/dashboard

AI Pre-Consultation Copilot

Before the appointment, AI synthesizes symptoms + reports into a structured clinical brief the doctor reviews before the patient walks in.

🗣️ Accepts symptoms in Hindi, Marathi, English

📊 Correlates with uploaded report data

🔔 Assigns urgency: LOW → CRITICAL

🎯 Suggests focus areas + recommended specialty

📋 Flags abnormal lab values with normal ranges

📈 Analysis reliability indicator (HIGH / MEDIUM / LOW)

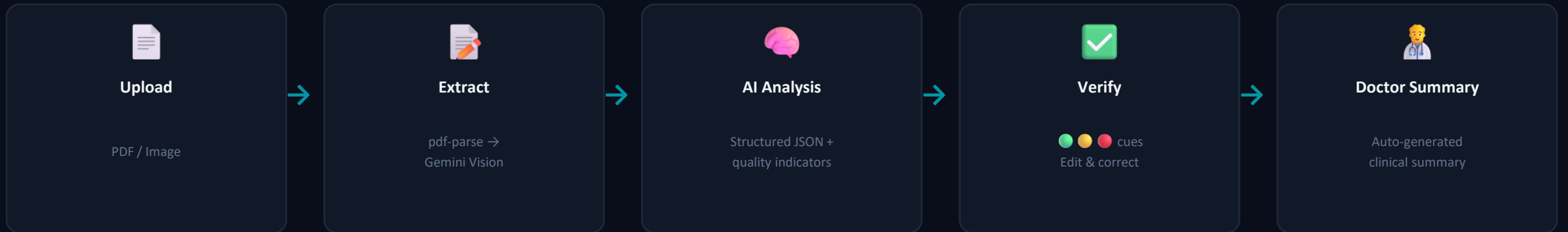
⚠️ Advisory output only — not a clinical diagnosis. Always reviewed by the consulting physician.

```
{
  "summaryText": "Patient reports fever and
    headache for 3 days. Lab reports
    indicate low hemoglobin.",
  "symptoms": ["fever", "headache", "fatigue"],
  "symptomTimeline": "3 days",

  "urgencyLevel": "MEDIUM",
  "conditions": ["Anemia", "Viral fever"],

  "suggestedFocusAreas": [
    "Complete blood count",
    "Iron studies"
  ],
  "abnormalValues": [{
    "parameter": "Hemoglobin",
    "value": "8.2 g/dL",
    "normalRange": "12.0–17.5 g/dL"
  }],
  "analysisReliability": "HIGH",
  "recommendedSpecialty": "General Physician"
}
```

Medical Document Intelligence



HIGH — AI Self-Assessment

Score ≥ 0.8 — AI reports high extraction certainty

MEDIUM — Verify Recommended

Score 0.5–0.8 — patient should double-check this value

LOW — Manual Check Needed

Score < 0.5 — patient should enter correct value

Honest note: These indicators are AI-generated, not clinically validated. They serve as attention signals — helping patients identify which values to double-check. Patient corrections are tracked for future model improvement.

 Screenshot: MedicalVerificationModal with green/amber/red indicators

Multilingual Voice-First Input

Patients describe symptoms by speaking naturally — in their own language.

GB English

IN हिंदी

IN मराठी



Real-time live transcription (browser-native speech recognition)



AI mirrors input language in responses



Urgency keywords (chest pain, breathlessness) trigger instant alerts



Quick-pick symptom tags for common conditions



Hinglish (mixed Hindi-English) handled naturally



Voice Input (Hindi):

"मुझे 3 दिन से बुखार और सिरदर्द है,
और सीने में दर्द हो रहा है"

⚠ Urgent symptom detected: chest pain — call 112

AI normalizes for doctor's brief:

→ fever, headache, chest pain (3-day duration)

AI-Powered Patient 360°

When the doctor opens a patient profile, they see an AI Command Center — not a blank chart.



AI Patient Snapshot

Risk badge (LOW→CRITICAL), current symptoms, conditions, allergies — one hero view



Consultation Prep Panel

AI-suggested focus areas, new symptoms since last visit, recent abnormal findings



Health Trends

Biomarker trend charts (hemoglobin, glucose, cholesterol) with SVG line charts



Health Timeline

Chronological patient history — appointments, reports, risk levels



Reports Viewer

View PDFs and images inline. See AI analysis severity and extraction data.



Emergency Triage

4-level AI severity classification (CRITICAL/HIGH/MODERATE/LOW) and 112 quick-dial



Screenshot: PatientView360 — AI Snapshot Hero with risk badge, symptoms, conditions

Technical Foundation

Frontend

React 18 + Vite
Tailwind CSS + Framer Motion
Zustand state management
Browser-native voice API
Lazy-loaded routes

Backend

Node.js + Express.js
Modular service layer
JWT + OTP authentication
Cron-based reminders
Role-based access

AI & Data

Google Gemini API
4-model fallback chain
MongoDB Atlas
Cloudinary CDN
Nodemailer

AI Resilience

4-tier model fallback: gemini-flash-latest → 2.0-flash-lite → 2.0-flash → 2.5-flash. Daily quota exhaustion cached. Safety blocks throw immediately.

Security

Helmet.js headers, MongoDB injection sanitization, CORS whitelist, role-based rate limiting (auth: 20/15min, AI: 20/15min, uploads: 10/15min), medical disclaimers.

Competitive Landscape

AI Capability	Practo	Ada Health	Nuance DAX	ArogyaAI
Appointment Booking	✓	—	—	✓
AI Symptom Chat	—	✓	—	✓
Medical Document AI	—	—	—	✓
AI Quality Indicators	—	—	—	✓
Human-in-the-Loop	—	—	—	✓
Pre-Consultation Copilot	—	—	Post-visit	✓
Multilingual Voice	—	—	EN only	✓
AI Emergency Triage	—	✓	—	✓
Total AI Features	1/8	2/8	1/8	8/8

⚡ Honest note: This comparison focuses on AI capabilities. Practo has millions of users, real payments, and verified doctors — operational capabilities we haven't built yet. Our advantage is in AI intelligence, not operational scale.

Who Benefits — And Why We Can Win

For Patients

-  Accessibility: Speak symptoms in your language
-  Transparency: See AI extractions, correct mistakes
-  Continuity: Health records organized
-  Safety: Emergency triage when seconds matter

For Doctors

-  Pre-read Brief: AI copilot before consultation
-  Time Efficiency: 1 min saved × 40+ patients
-  Structured Data: Abnormal values flagged
-  Automation: Cron reminders reduce no-shows

Responsible AI

-  Advisory only — not a diagnosis
-  Human verification for extracted data
-  Doctors remain final decision makers
-  Medical disclaimers on every AI output

Working Prototype

Running code, not slides

5 AI Capabilities

Chat, OCR, copilot, triage, summary

3 Languages

Hindi, Marathi, English

Deployment-Ready

Fallback chain, auth, rate limiting

1.4B People

Solving a real problem at scale

Future Scope & Validation Roadmap

Future Development Phases

PHASE 1

NOT YET IMPLEMENTED

Verified doctor onboarding · Razorpay payments · Video consultations

PHASE 2

NOT YET IMPLEMENTED

ABDM / ABHA integration · Digital prescriptions · Unified health records

PHASE 3

NOT YET IMPLEMENTED

Chronic disease prediction · Longitudinal monitoring · AI visit summaries

PHASE 4

NOT YET IMPLEMENTED

Android & iOS apps · Tamil, Telugu, Bengali, Kannada

PHASE 5

NOT YET IMPLEMENTED

Wearable integrations · Remote patient monitoring

Validation & Ethics Roadmap



Clinical validation with practicing doctors



Real-world pilot testing in healthcare facilities



Accuracy benchmarking against ground truth data



Explainability improvements for AI decisions



Bias and safety evaluation across demographics



Regulatory readiness assessment (DISHA, CDSCO)



All validation items are future work. We know the path from prototype to product — and we've designed the architecture to support it.

Thank You

In India, a doctor sees 40 patients a day.

Each consultation lasts 2 minutes.

We can't add more minutes.

But we can make every minute count.



GitHub: ArogyaAI



Live Demo Available

Questions? We'd love to discuss the technical details — or show a live demo.